



CLUSTERING

INTRODUCTION

Unsupervised learning is a type of machine learning that learns from **test data** that **has not been labeled**, classified or categorized.

Unsupervised learning subsumes all kinds of machine learning **where there is no known output, no teacher to instruct the learning algorithm.**

In unsupervised learning, the learning algorithm is **just shown the input data** and **asked to extract knowledge** from this data.

INTRODUCTION

Instead of responding to feedback, unsupervised learning **identifies commonalities in the data** and **reacts based on the presence or absence of such commonalities** in each new piece of data.

Unsupervised learning is used in **descriptive modelling**.

INTRODUCTION

There are many kinds of unsupervised learning such as:

- **Clustering**

- Clustering is grouping the data instances into predetermined number of groups based on the similarity between these instances.

- **Anomaly Detection**

- (or outlier detection) is the identification of the rare or unusual items, events or observations which raise suspicions by differing significantly from the majority of the data.

- **Association Rules**

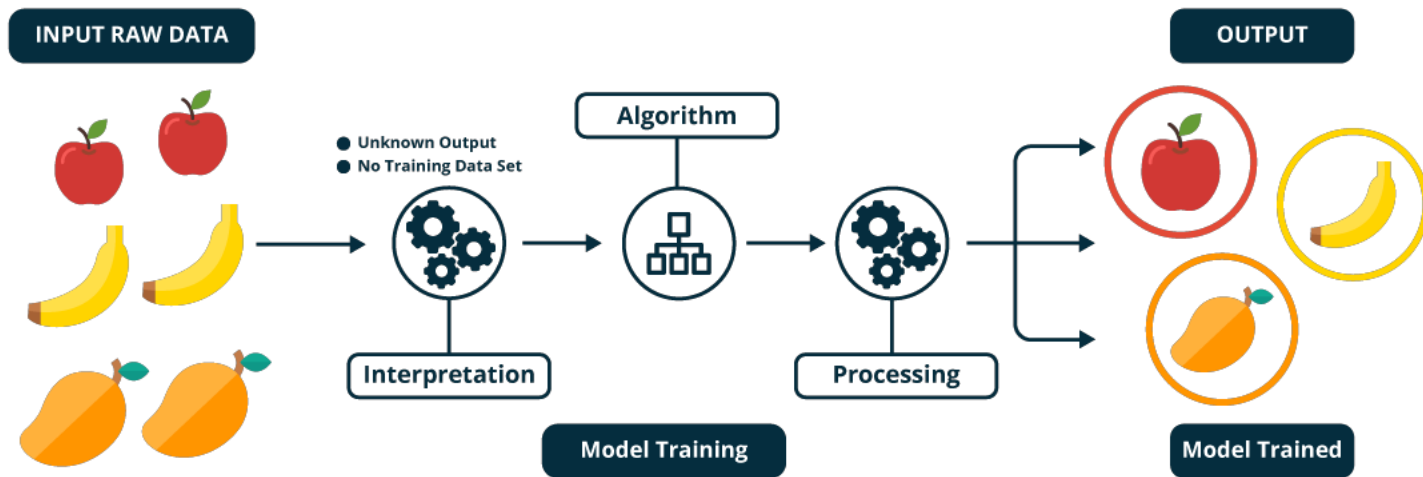
- Association Rules refers to the process to find the association between features' values in the data set to make decision based on these rules later.

- **Transformations**

- Transformations refers to the process to create a new easy/simple representation of the dataset to simplify the dataset understanding and the learning process as well.

CLUSTERING

Clustering algorithms **partition** data into **distinct groups** of **similar items** called **clusters**.



CLUSTERING

Clustering is a method of *grouping data that share similar trend and patterns*.

The **goal** is to split up the data in such a way that **points within a single cluster** are **very similar** and points in different clusters are different.

Similarly to classification algorithms, clustering algorithms assign (or predict) **a number to each data point**, indicating which cluster a particular point belongs to.

CLUSTERING

BASIC CONCEPTS

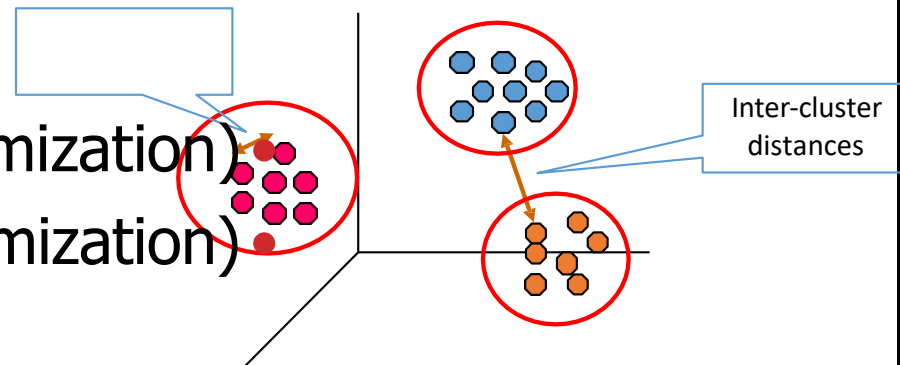
Given Points (or objects) in some space – often a high dimensional space – group the points into small number of clusters, **each cluster consisting of points that are “near” in some sense.**

similar (or related) to one another within the same group and **dissimilar** (or unrelated) to the objects in other groups.

or
Intra-cluster
distances

Inter-clusters distance (maximization)

Intra-clusters distance (minimization)



CLUSTERING

EXAMPLE: CUSTOMERS' PURCHASES

Customers in **cluster 1**
purchase few high-priced items

- Rows: 1, 2 & 3

Customers in **cluster 2**
purchase many high-price
items

- Rows: 4, 5 & 6

Customer in **cluster 3** purchase
few low-priced items.

- Rows: 7, 8 & 9

Row#	# of items	Price
1	2	1700
2	3	2000
3	4	2300
4	10	1800
5	12	2100
6	11	2500
7	2	100
8	3	200
9	3	350

CLUSTERING

POSSIBLE APPLICATIONS

Clustering can be used in different applications such as:

- **Targeted Marketing** to discover customer groups.
- **Genomics** to find groups of gene with similar expressions.
- **Astronomy** to find groups of similar stars and galaxies.
- **Insurance**: Identifying groups of motor insurance policy holders with a high average claim cost
- **City-planning**: Identifying groups of houses according to their house type, value, and geographical location

CLUSTERING

CLUSTER CAN BE AMBIGUOUS



How many clusters?



Two Clusters



Four Clusters



Six Clusters

CLUSTERING

TYPES OF CLUSTERING

A clustering is a set of clusters

- Important distinction between hierarchical and partitional sets of clusters

Partitional Clustering

- A division data objects into non-overlapping subsets (clusters) such that each data object exists in exactly one subset

Hierarchical clustering

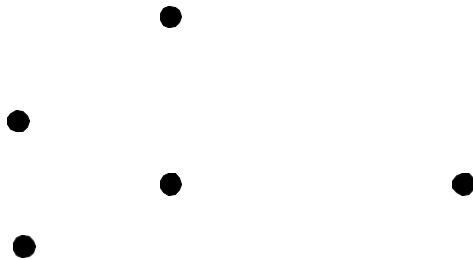
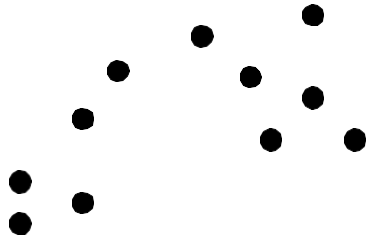
- A set of nested clusters organized as a hierarchical tree

PARTITIONAL CLUSTERING

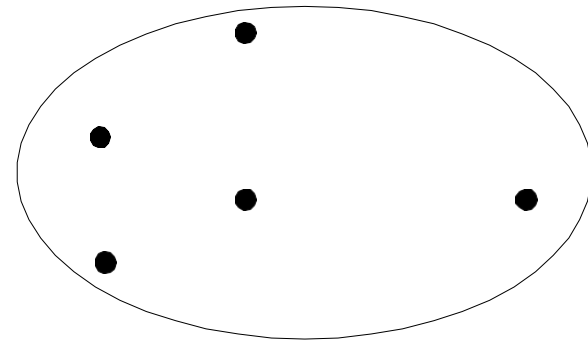
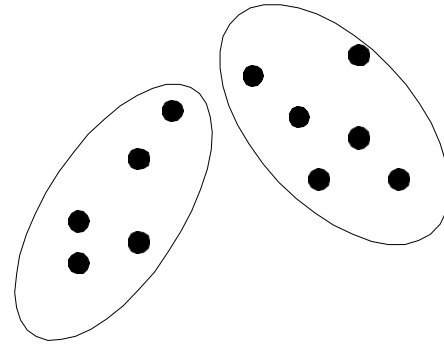
CLUSTERING

PARTITIONAL

CLUSTERING



Original Points

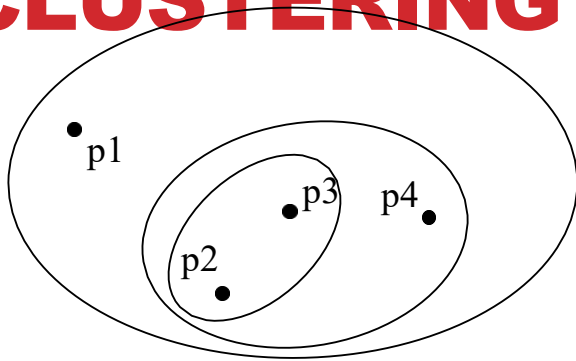


A Partitional Clustering

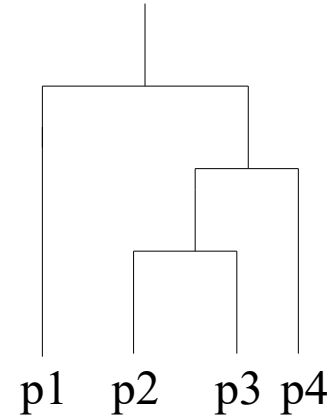
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HIERARCHICAL

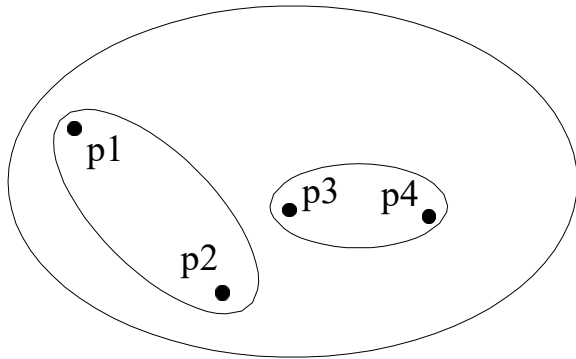
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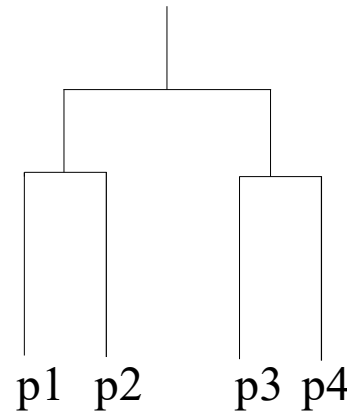
Traditional Hierarchical Clustering



Traditional Dendrogram



Non-traditional Hierarchical Clustering



Non-traditional Dendrogram

CLUSTERING

CLUSTERING VS CLASSIFICATION

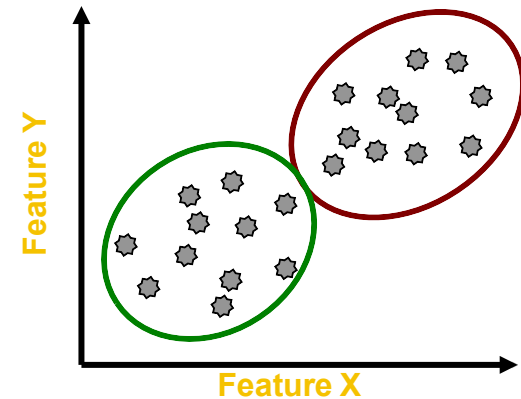
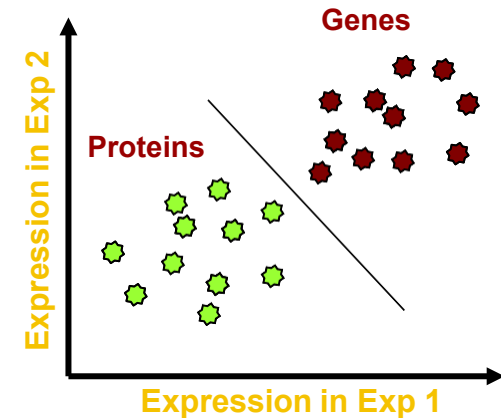
Objects characterized by one or more features

Classification

- Have labels for some points
- Want a “rule” that will accurately assign labels to new points
- Supervised learning

Clustering

- No labels
- Group points into clusters based on how “near” they are to one another
- Identify structure in data
- Unsupervised learning



PARTITIONAL CLUSTERING

A clustering algorithm attempts to find natural groups of components (or data) based on some similarity. Also, the clustering algorithm finds the centroid of a group of data sets.

To determine cluster membership, most algorithms evaluate the distance between a point and the cluster centroids.

The algorithm clustering a from a basically is cluster the of description statistical the with centroids number of components in each cluster.

PARTITIONAL CLUSTERING

K-MEANS CLUSTERING

k-means clustering is one of the simplest and most commonly used clustering algorithms.

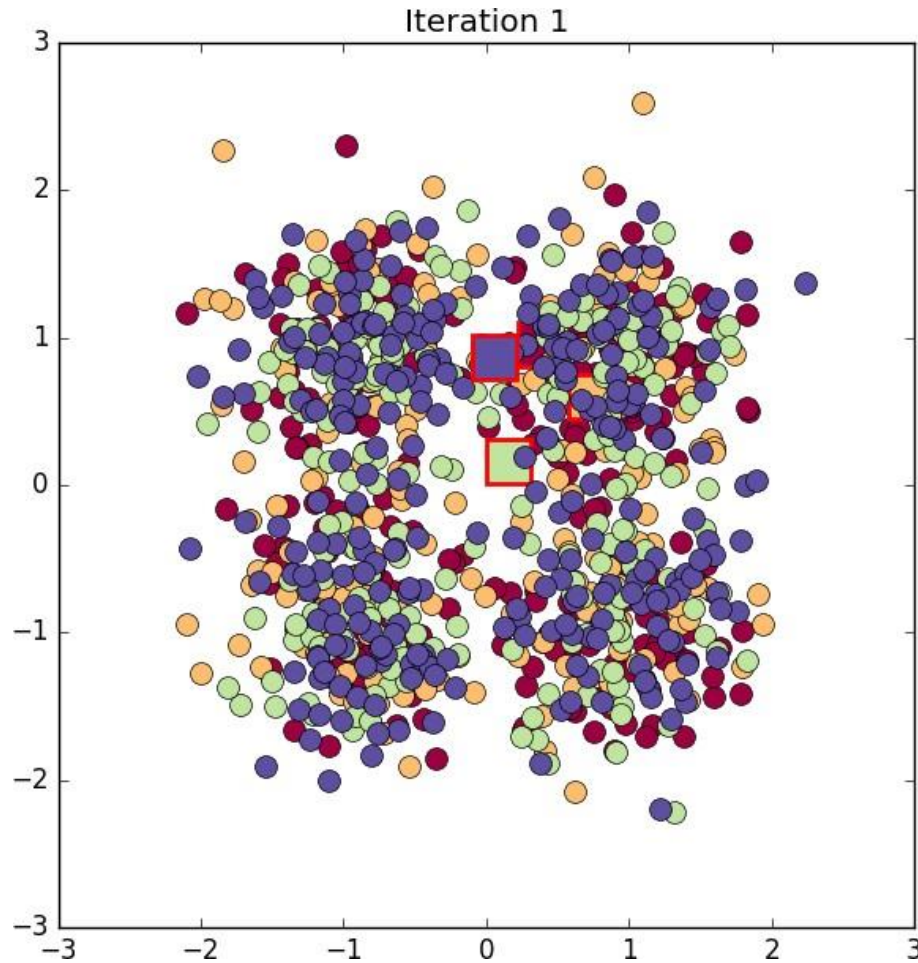
It tries to find **cluster centers** that are **representative of certain regions** of the data.

The algorithm **alternates** between **two steps**:

- assigning each data point to the closest cluster center, and then setting each cluster center as the mean of the data points that are assigned to it.
 - **Distance metric** is used to assign the point to closest center
 - E.g.: **Euclidian** distance or **Manhattan** distance
- The algorithm is finished when the assignment of instances to clusters no longer changes.

- Jaccard
- Spectral
- n
- Hammin
- Cosine

PARTITIONAL CLUSTERING K-MEANS CLUSTERING



PARTITIONAL CLUSTERING K-MEANS CLUSTERING

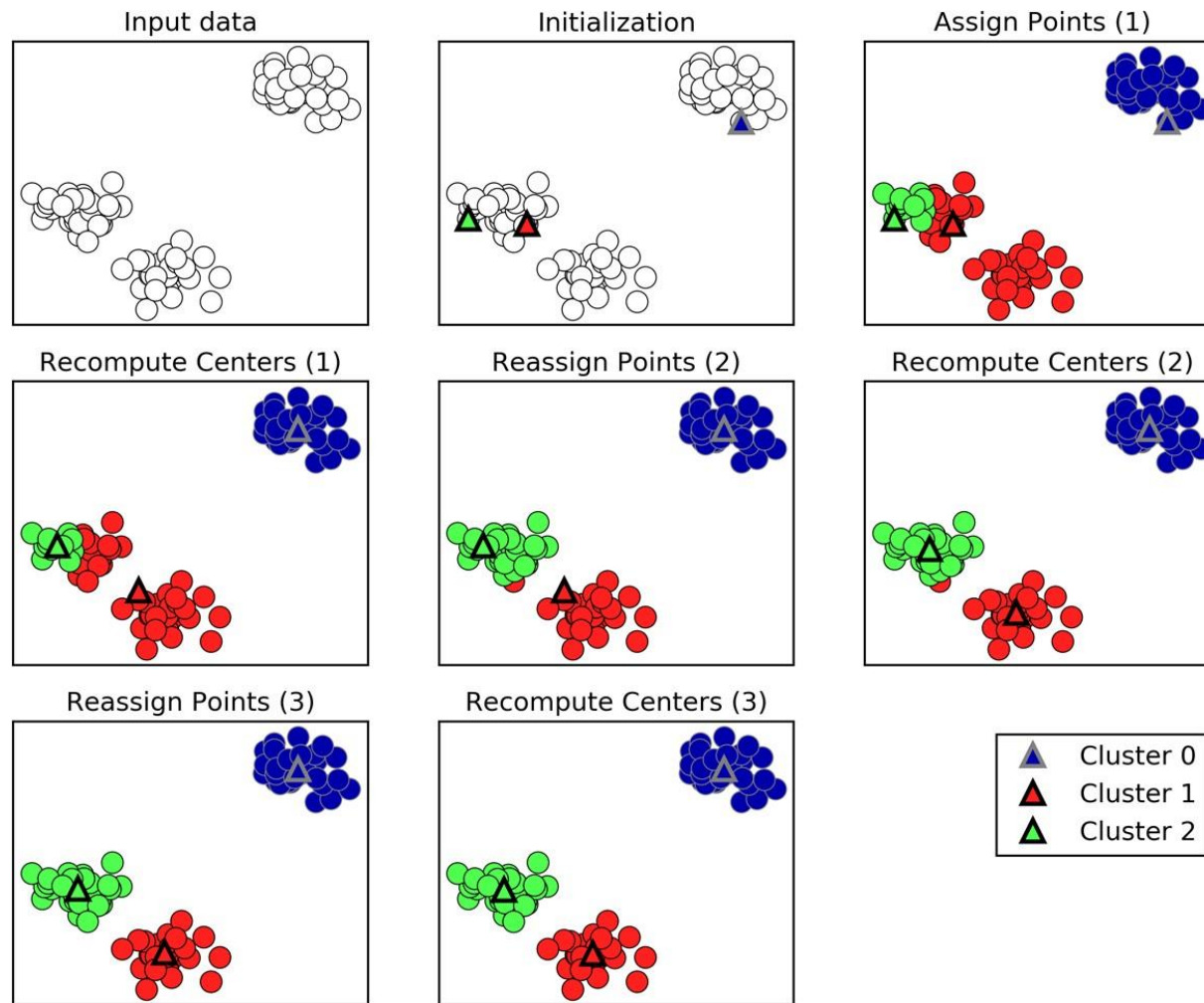


Figure 3-23. Input data and three steps of the k-means algorithm

PARTITIONAL CLUSTERING

K-MEANS CLUSTERING

on depends algorithm clustering k-mean of success The
providing it with the right number of clusters.

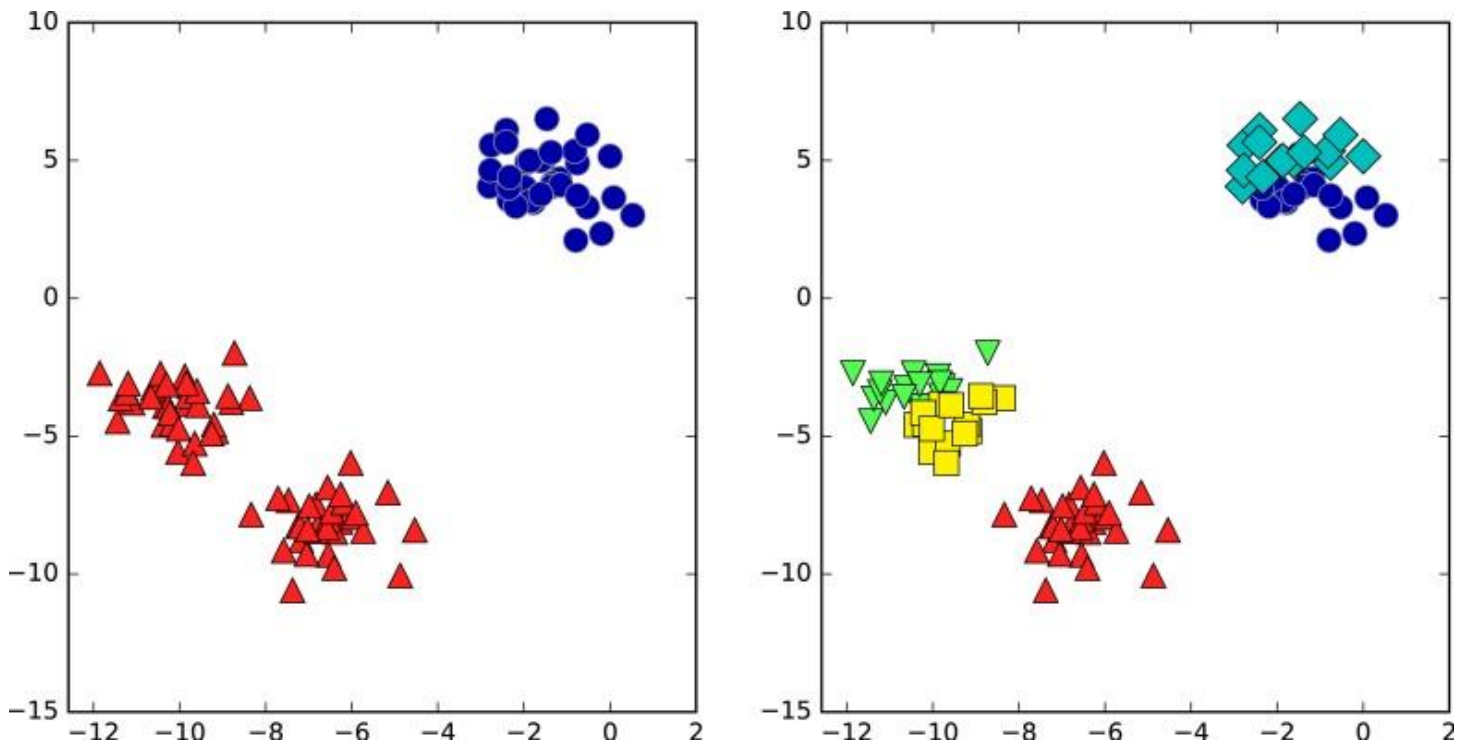


Figure 3-26. Cluster assignments found by k-means using two clusters (left) and five clusters (right)

PARTITIONAL CLUSTERING K-MEAN EXAMPLE 1

Given: $\{2,4,10,12,3,20,30,11,25\}$, $k=2$

1. Randomly assign two centers: $C_1=3$, $C_2=4$
2. $K_1=\{2,3\}$, $K_2=\{4,10,12,20,30,11,25\}$, $C_1=2.5$, $C_2=16$
3. $K_1=\{2,3,4\}$, $K_2=\{10,12,20,30,11,25\}$, $C_1=3$, $C_2=18$
4. $K_1=\{2,3,4,10\}$, $K_2=\{12,20,30,11,25\}$, $C_1=4.75$, $C_2=19.6$
5. $K_1=\{2,3,4,10,11,12\}$, $K_2=\{20,30,25\}$, $C_1=7$, $C_2=25$

Stop as the clusters with these means are the same.

PARTITIONAL CLUSTERING

K-MEAN EXAMPLE 2

Cluster the following eight points (with (x, y) representing locations) **into three clusters**.

$A_1(2, 10)$ $A_2(2, 5)$ $A_3(8, 4)$ $A_4(5, 8)$ $A_5(7, 5)$ $A_6(6, 4)$
 $A_7(1, 2)$ $A_8(4, 9)$

Initial **cluster centers** are: $A_1(2, 10)$, $A_4(5, 8)$ and $A_7(1, 2)$.
The **distance function** between two points $a=(x_1, y_1)$ and $b=(x_2, y_2)$ is defined as: $d(a, b) = |x_2 - x_1| + |y_2 - y_1|$.

Use k-means algorithm to find the three cluster centers after the second iteration.

PARTITIONAL CLUSTERING

K-MEAN EXAMPLE 2

First Iteration:		Centers			
		C1 (2, 10)	C2 (5, 8)	C3 (1, 2)	
Point		Dist from C1	Dist from C2	Dist from C3	Cluster (C?)
A1	(2, 10)	0	5	9	1
A2	(2, 5)	5	6	4	3
A3	(8, 4)	12	7	9	2
A4	(5, 8)	5	0	10	2
A5	(7, 5)	10	5	9	2
A6	(6, 4)	10	5	7	2
A7	(1, 2)	9	10	0	3
A8	(4, 9)	3	2	10	2

PARTITIONAL CLUSTERING

K-MEAN EXAMPLE 2

Sec	ond Itera	tion: Centers			
		(2, 10)	(6, 6)	(1.5, 3.5)	
Point		Dist from C1	Dist from C2	Dist from C3	Cluster (C?)
A1	(2, 10)	0	8	7	1
A2	(2, 5)	5	5	2	3
A3	(8, 4)	12	4	7	2
A4	(5, 8)	5	3	8	2
A5	(7, 5)	10	2	7	2
A6	(6, 4)	10	2	5	2
A7	(1, 2)	9	9	2	3
A8	(4, 9)	3	5	8	1

PARTITIONAL CLUSTERING

K-MEAN EXAMPLE 2

Th i	rd Iteration	n: Centers			
		C1 (3,9.5)	C2 (6.5,5.2)	C3 (1.5,3.5)	
Point		Dist from C1	Dist from C2	Dist from C3	Cluster (C?)
A1	(2, 10)	1.5	9.3	7	1
A2	(2, 5)	5.5	4.7	2	2
A3	(8, 4)	10.5	2.7	7	2
A4	(5, 8)	3.5	4.3	8	1
A5	(7, 5)	8.5	0.7	7	2
A6	(6, 4)	8.5	1.7	5	2
A7	(1, 2)	9.5	8.7	2	3
A8	(4, 9)	1.5	6.3	8	1

PARTITIONAL CLUSTERING

K-MEAN EXAMPLE 3

Assume that we have dataset with 8 point $\{(2, 10), (2, 5), (8, 4), (5, 8), (7, 5), (6, 4), (1, 2), (4, 9)\}$. Group it into different clusters depending on Manhattan distance.

	C1 (2, 5)	C2 (5, 8)	C3 (1, 2)	
Point	Dist from C1	Dist from C2	Dist from C3	Cluster
(2, 5)				
(2, 10)				
(8, 4)				
(5, 8)				
(7, 5)				
(6, 4)				
(1, 2)				
(4, 9)				

PARTITIONAL CLUSTERING: K-MEDOIDS

k-medoids – **instead of mean, use median**
of each cluster

Mean of 1, 3, 5, 7, 9 is • 5

Mean of 1, 3, 5, 7, 1009 is • 205

Median of 1, 3, 5, 7, 1009 is • 5

Median advantage: not affected by extreme values •

PARTITIONAL

CLUSTERING: BINARY VARIABLES

If the variables are **binary** such as patient is smoker (1) or non-smoker (0), how to compute the similarity.

In this case we compute the *dissimilarity*.

In general $d(i, j)$ is a **non-negative number** that is close to 0 when the objects i and j are highly similar or “near” each other.

CLUSTERING: BINARY VARIABLES

A contingency table for binary data

		Object j		
		1	0	<i>sum</i>
Object i	1	a	b	$a + b$
	0	c	d	$c + d$
	<i>sum</i>	$a + c$	$b + d$	p

$$d(i, j) = \frac{b + c}{a + b + c}$$

Jaccard coefficient of dissimilarity:

PARTITIONAL CLUSTERING: BINARY VARIABLES

Patient records are given below where (Y) and (P) means 1 and (N) means 0

Name	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	Y	N	P	N	N	N
Mary	Y	N	P	N	P	N
Jim	Y	Y	N	N	N	N

$$d(jack, mary) = \frac{(0 + 1)}{(2 + 0 + 1)} = 0.33$$

$$d(Jack, jim) = \frac{(1 + 1)}{(1 + 1 + 1)} = 0.67$$

$$= d(jim, mary) = \frac{(1 + 2)}{(1 + 1 + 2)} = 0.75$$

	Mary		
	1	0	sum
Jack	1	2	0
	0	1	3
sum	3	3	6

PARTITIONAL

CLUSTERING: BINARY VARIABLES

$$\text{dissimilarity}1 - \text{Similarity}$$

These results suggested that *Jim* and *marry* unlikely to have similar disease and *jack* and *marry* likely to have a similar disease.

For categorical variables , convert categorical to binary and use the same method

PARTITIONAL CLUSTERING

K-MEANS CLUSTERING

Advantages

Simple,
understandable
items automatically
assigned to clusters.

Disadvantages

of numberpickMust
clusters before hand
All items forced into a
cluster
Sensitive to outliers
initialto sensitive
seeds
It is not suitable for
discovering clusters
that are not hyper-
ellipsoids (or hyper-
spheres).

CLUSTERING

FAILURE CASES OF K-MEANS

Even if you know the “right” number of clusters for a given dataset, k -means **might not** always **be able to recover** them.

Each cluster is **defined only by its center**, which means that each cluster is a convex, arched, shape.

As a result of this, k -means can only capture relatively simple • shapes.

k -means also assumes that all clusters have the same “diameter” in some sense; it always draws the boundary between clusters to be exactly in the middle between the cluster centers.

CLUSTERING

FAILURE CASES OF K-MEANS

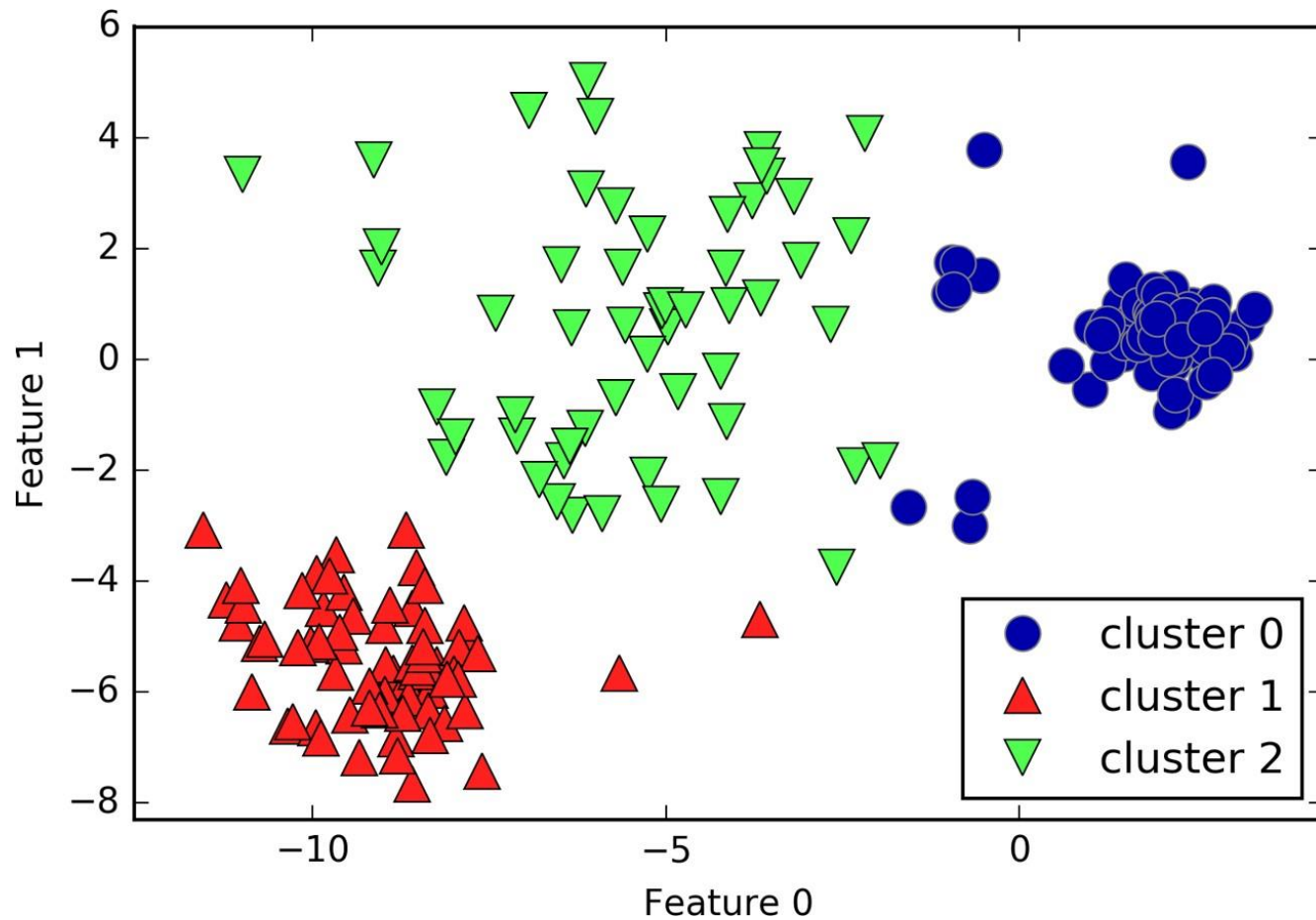


Figure 3-27. Cluster assignments found by k-means when clusters have different densities

CLUSTERING

FAILURE CASES OF K-MEANS

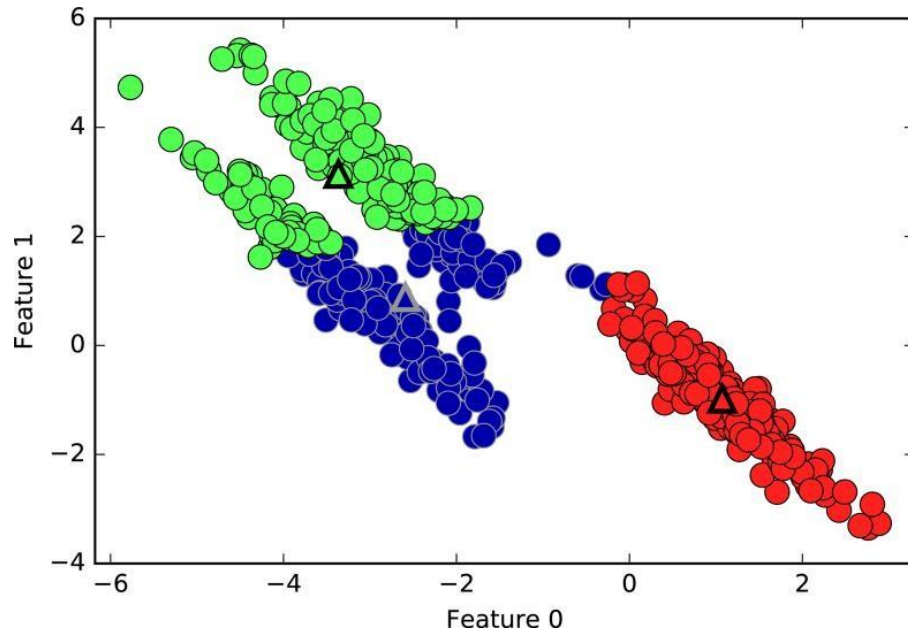


Figure 3-28. *k-means* fails to identify nonspherical clusters

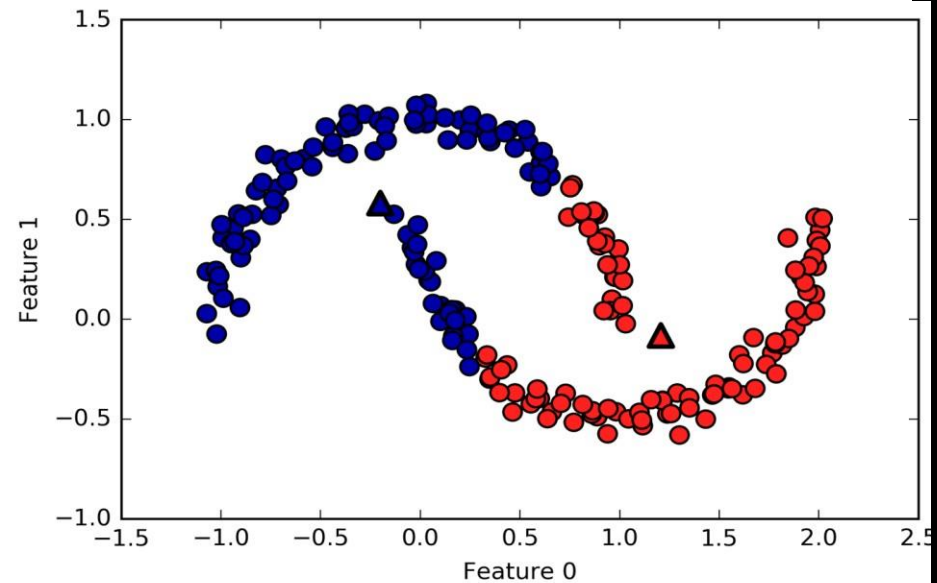


Figure 3-29. *k-means* fails to identify clusters with complex shapes

HIERARCHICAL CLUSTERING

AGGLOMERATIVE CLUSTERING

clustering of collections refers to clustering algorithms that all build upon the same principles:

- the algorithm starts by declaring each point its own cluster, and then merges the two most similar clusters until some stopping criterion is satisfied.

- The most used stopping criterion is the number of clusters, so similar clusters are merged until only the specified number of clusters are left.

- The linkage criterion determines which distance to use between sets of observation. The clustering algorithm will merge the pairs of cluster that minimize this criterion.

- There are several *linkage criteria* that specify how exactly the "most similar cluster" is measured. This measure is always defined between two existing clusters.

AGGLOMERATIVE CLUSTERING

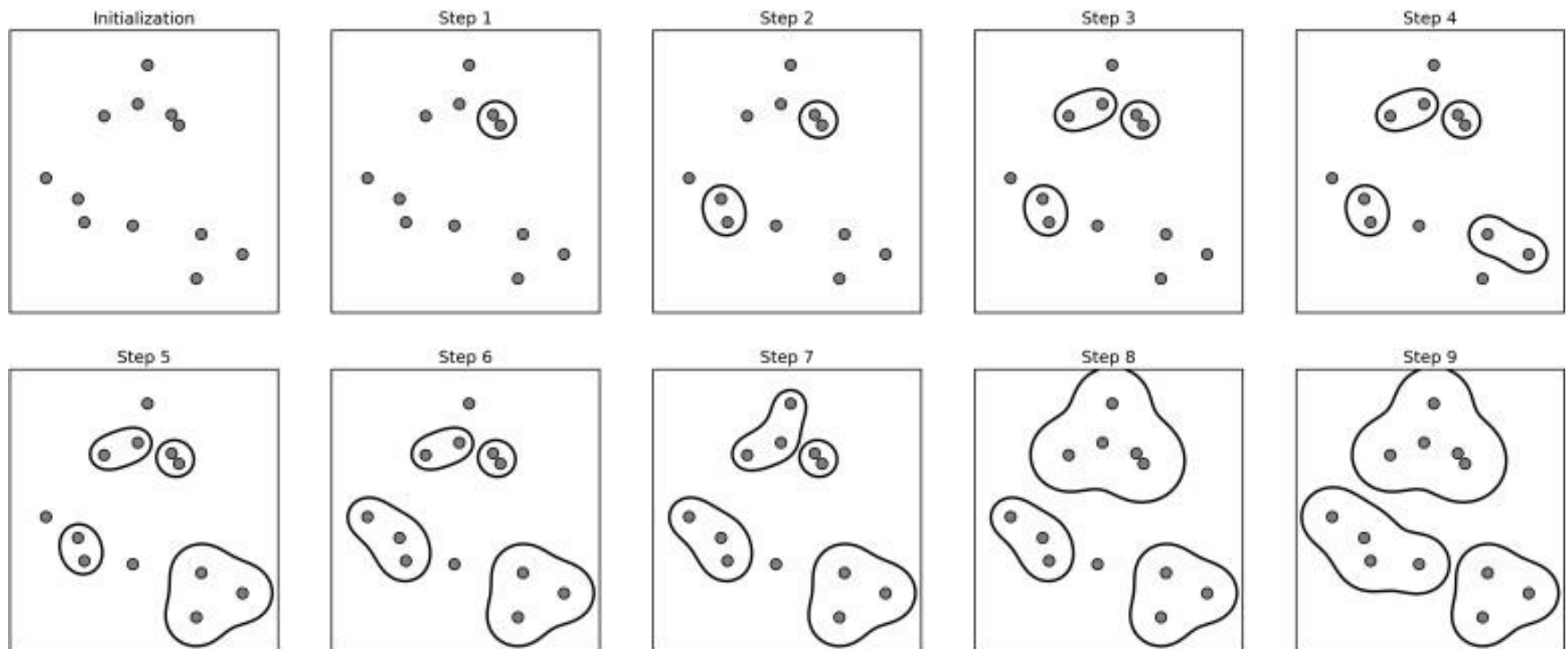


Figure 3-33. Agglomerative clustering iteratively joins the two closest clusters

AGGLOMERATIVE CLUSTERING

Some *linkage criteria* that:

- **Ward:** The default choice, ward picks the two clusters to merge such that **the variance within all clusters increases the least.** This leads to the formation of clusters that are as compact and equally sized as possible.
- **Single:** It combines the two clusters that contain the closest pair of elements not yet belonging to the same cluster as each other.
- **Average:** merges the two clusters that have the smallest average distance between all their points.
- **Complete:** also known as maximum linkage, merges the two clusters that have the smallest maximum distance between their points.

HIERARCHICAL CLUSTERING AND DENDROGRAMS

produces clustering Agglomerative a as knowniswhat
hierarchical clustering.

The clustering proceeds iteratively, and every point makes a journey from being a single point cluster to belonging to some final cluster.

Each intermediate step provides a clustering of the data (with a different number of clusters). It is sometimes helpful to look at all possible clusterings jointly.

HIERARCHICAL CLUSTERING AND DENDROGRAMS

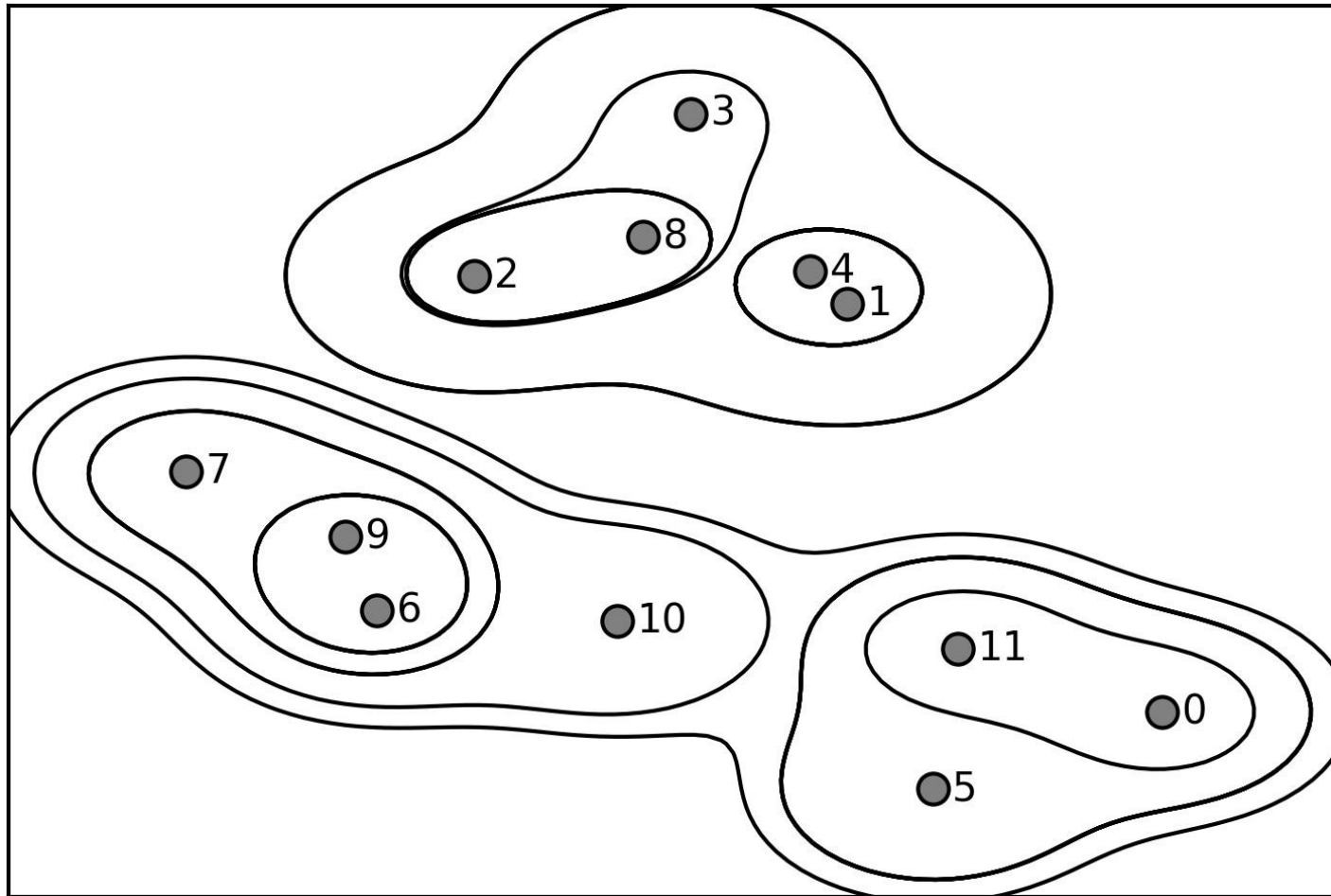


Figure 3-35. Hierarchical cluster assignment (shown as lines) generated with agglomerative clustering, with numbered data points (cf. [Figure 3-36](#))

HIERARCHICAL CLUSTERING AND DENDROGRAMS

While this visualization provides a **very detailed view** of the hierarchical clustering, it **relies on the two-dimensional nature of the data** and therefore **cannot be used** on datasets **that have more than two features.**

is, There is another tool, however, called *dendrogram*, that can handle multidimensional datasets.

HIERARCHICAL CLUSTERING AND DENDROGRAMS

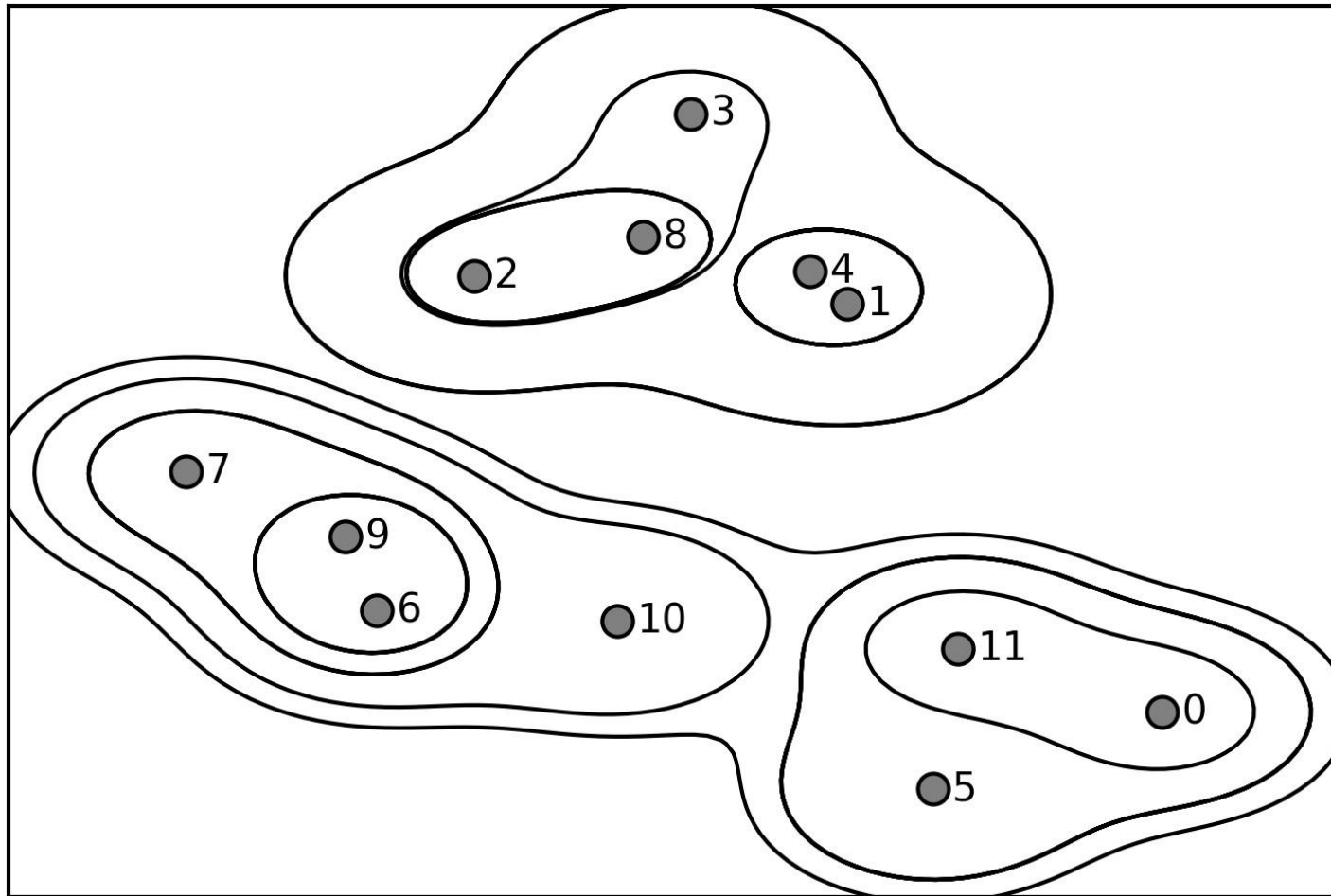


Figure 3-35. Hierarchical cluster assignment (shown as lines) generated with agglomerative clustering, with numbered data points (cf. [Figure 3-36](#))

HIERARCHICAL CLUSTERING AND DENDROGRAMS

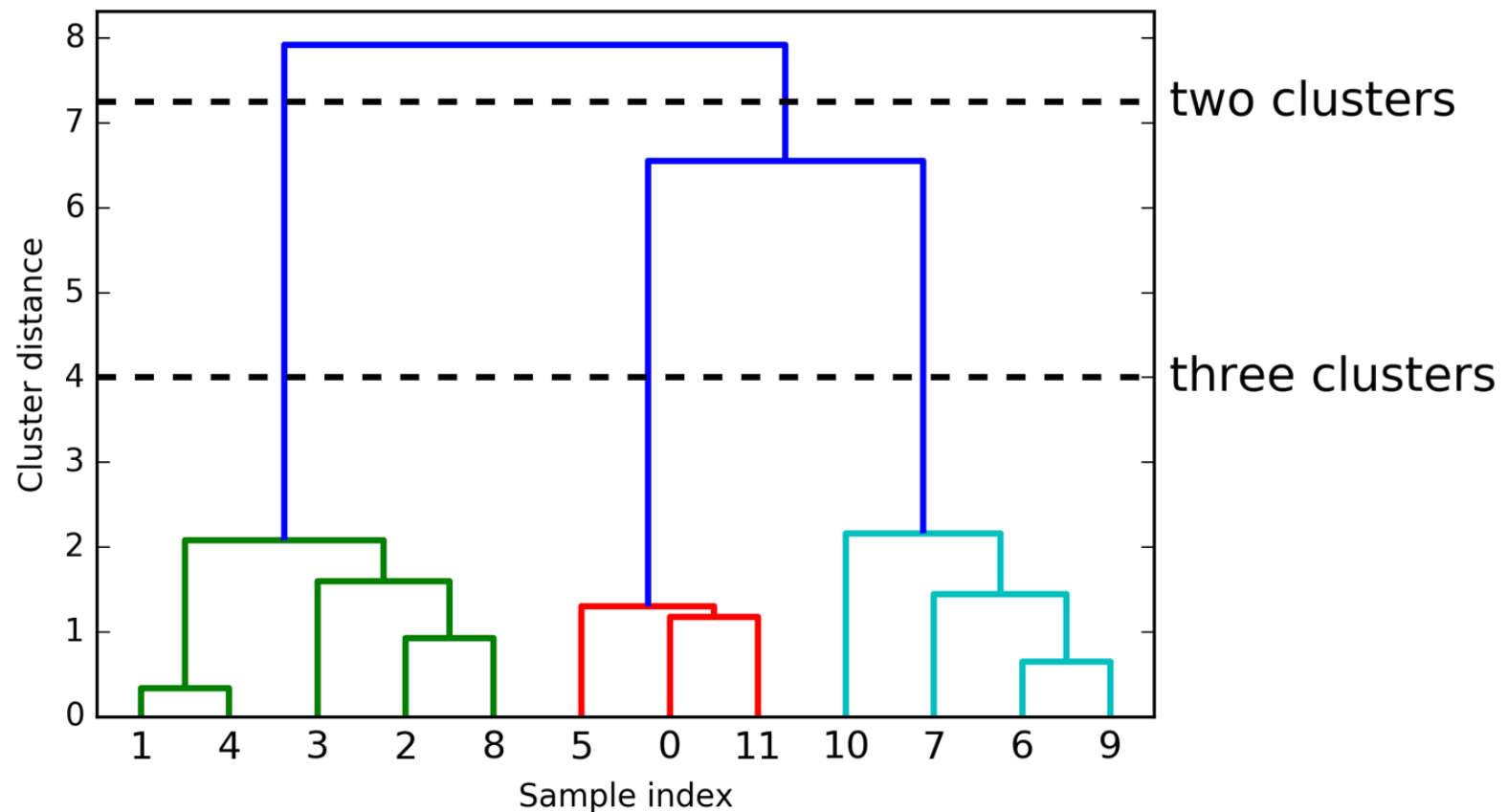


Figure 3-36. Dendrogram of the clustering shown in Figure 3-35 with lines indicating splits into two and three clusters

HIERARCHICAL CLUSTERING

EXAMPLE

Hierarchical clustering of distances between cities in

kilometers between some Italian cities.

The used method is *single-linkage*.



	BA	FI	MI	NA	RM
BA	0	662	877	255	412
FI	662	0	295	468	268
MI	877	295	0	754	564
NA	255	468	754	0	219
RM	412	268	564	219	0
TO	966	400	138	869	669

HIERARCHICAL

CLUSTERING EXAMPLE 1...

The **nearest pair of cities** is MI and TO, at distance 138. These are merged into a single cluster called "MI/TO".

The **level of the new cluster** is $L(\text{MI/TO}) = 138$ and the **new sequence number** is $m = 1$.

Then we compute the distance from this new compound object to all other objects

In **single-linkage** clustering the rule is that the distance •
from the compound object to another object is equal to

HIERARCHICAL CLUSTERING: EXAMPLE...

So the distance from "MI/TO" to RM is chosen to be 564, which is the distance from MI to RM, and so on.

After merging MI with TO we obtain the following

	BA	FI	MI/TO	NA	RM
BA	0	662	877	255	412
FI	662	0	295	468	268
MI/TO	877	295	0	754	564
NA	255	468	754	0	219
RM	412	268	564	219	0

HIERARCHICAL CLUSTERING: EXAMPLE...

$$\min d(i, j) = d(\text{NA}, \text{RM}) = 219,$$

then merge NA and RM into a new cluster

called NA/RM •

$$L(\text{NA/RM}) = 219$$

	BA	FI	MI/TO	NA/RM
BA	0	662	877	255
FI	662	0	295	268
MI/TO	877	295	0	564
NA/RM	255	268	564	0

HIERARCHICAL

CLUSTERING: EXAMPLE...

$$255, = d(\textcolor{red}{BA}, \textcolor{blue}{NA/RM}) = \min d(i, j)$$

then merge BA and NA/RM into a new cluster

called BA/NA/RM •

$$L(\text{BA/NA/RM}) = 255$$

m = 3

	BA NA/RM	FI	MI/TO
BA/NA/R M	0	268	564
FI	268	0	295
MI/TO	564	295	0

HIERARCHICAL CLUSTERING EXAMPLE...

$$d(\text{BA/NA/RM}, \text{FI}) = 268,$$

then merge BA/NA/RM and FI a new cluster

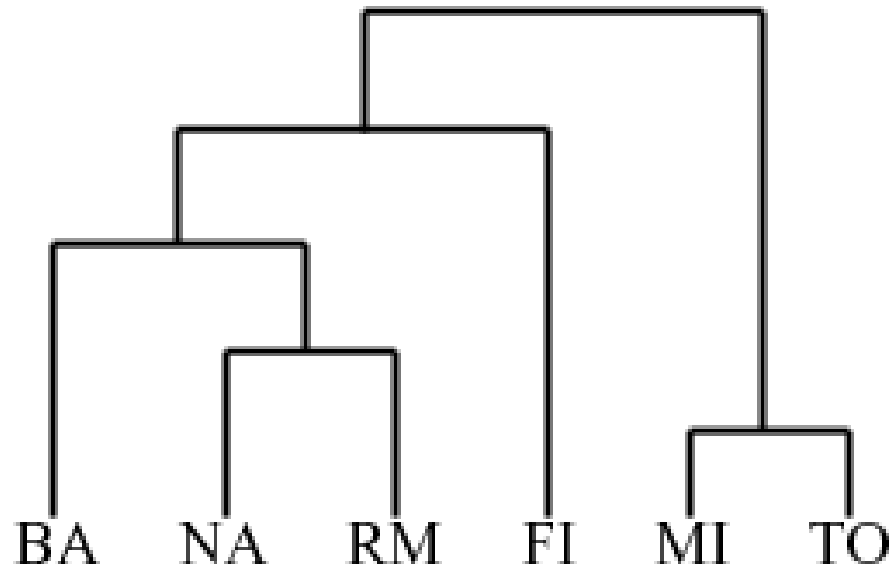
Called BA/FI/NA/RM • into
 $L(\text{BA/FI/NA/RM}) = 268$

$m = 4$

HIERARCHICAL

CLUSTERING EXAMPLE...

Finally, we merge the last two clusters at level 295.
The process is summarized by the following hierarchical tree



CLUSTERING EVALUATION

EVALUATING CLUSTERING ALGORITHMS

One of the challenges in applying clustering algorithms is that **it is very hard to assess how well an algorithm worked**, and to compare outcomes between different algorithms.

When applying clustering algorithm **on training set** we can compute the **purity** of each cluster, and **on test set** it is **almost impossible**.

	T1	T2	T3
C1	0	25	5
C2	15	0	20
C3	10	0	0

$$^1 \sum \max_{t \in C} = Purity$$

EVALUATING CLUSTERING ALGORITHMS

Purity Metric:

- Can be computed if we group the Test set, contains target class, using clustering algorithms.
- Assume that we have Test set of 24 elements belong into 3 different classes {○:8, □:8, ○:8}. The clustering algorithm gives 3 different clusters as follow:

$$C1 = \{\underset{9}{\underset{6}{\text{○, ○, □, ○, ○, ○, ○, ○, □}}}\} \quad \bullet \quad \text{then Purity of } C1 = \frac{\max(\text{○:1, ○:6, □:2})}{9}$$

$$C2 = \{\text{○, ○, ○, □, □, □, □, □}\} \quad \text{then Purity of } C2 = \frac{\max(\text{○:2, ○:1, □:5})}{8} = \bullet \frac{5}{8}$$

$$C3 = \{\text{○, ○, ○, ○, ○, □, ○}\} \quad \text{then Purity of } C3 = \frac{\max(\text{○:5, ○:1, □:1})}{7} = \bullet \frac{5}{7}$$

$$\text{The purity of the clustering algorithm with this dataset is} = \bullet \frac{6:5:5}{24} = \frac{16}{24} = 66.67\%$$

EVALUATING CLUSTERING ALGORITHMS

There are many metrics that can be used to assess the outcome of a clustering algorithm relative to a ground truth clustering.

- *ground truth clustering* means the algorithm will be evaluated based on the real classification of the data samples.
 - i.e. clustering algorithm will be applied on training set

The Adjusted Rand Index (ARI) 1
The most important ones •

The Normalized Mutual Information (NMI) 2

Both provide a quantitative measure between 0 and 1.

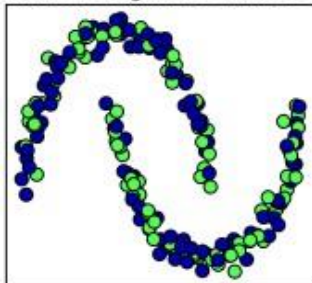
ALGORITHMS EVALUATING CLUSTERING WITH GROUND TRUTH

Adjusted Rand Index (ARI)

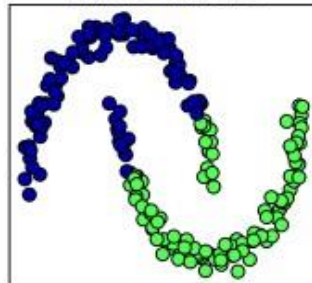
- The real name is Rand Index Adjusted for chance
- The **Rand Index (RI)** computes a **similarity** measure between **two clusters** by **considering all pairs of samples** and counting pairs that are assigned in the same or different clusters in the predicted and true clusters.

$$ARI = \frac{\sum_{ij} \binom{n_{ij}}{2} - [\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}] / \binom{n}{2}}{\frac{1}{2} [\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}] - [\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}] / \binom{n}{2}}$$

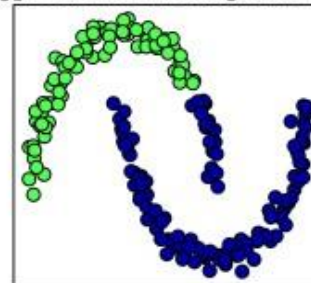
Random assignment - ARI: 0.00



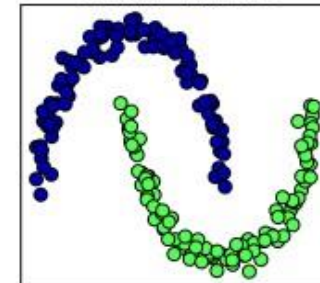
KMeans - ARI: 0.50



AgglomerativeClustering - ARI: 0.61



DBSCAN - ARI: 1.00



EVALUATING CLUSTERING ALGORITHMS

$$ARI = \frac{\sum_{ij} \binom{n_{ij}}{2} - [\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}] / \binom{n}{2}}{\frac{1}{2} [\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}] - [\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}] / \binom{n}{2}}$$

n_{ij} is across the diagonal (i.e., when $i = j$)

a_i is the row sums

b_j is the column sums

Let's assume we have the table:

y_i is an actual class & x_i is a predicted

	x1	x2	x3	Sums
y1	1	1	0	2
y2	1	2	1	4
y3	0	0	4	4
Sums	2	3	5	

Breaking into components:

- $\sum_{ij} \binom{n_{ij}}{2} = \binom{1}{2} + \binom{2}{2} + \binom{4}{2} = 7$
- $\sum_i \binom{a_i}{2} = \binom{2}{2} + \binom{4}{2} + \binom{4}{2} = 13$
- $\sum_j \binom{b_j}{2} = \binom{2}{2} + \binom{3}{2} + \binom{5}{2} = 14$

So, then

$$ARI = \frac{7 - 13 * 14 / 45}{(13 + 14) / 2 - 13 * 14 / 45} = 0.313$$

That is all for this chapter